

Asian Options

(1. Geometric Averaging)

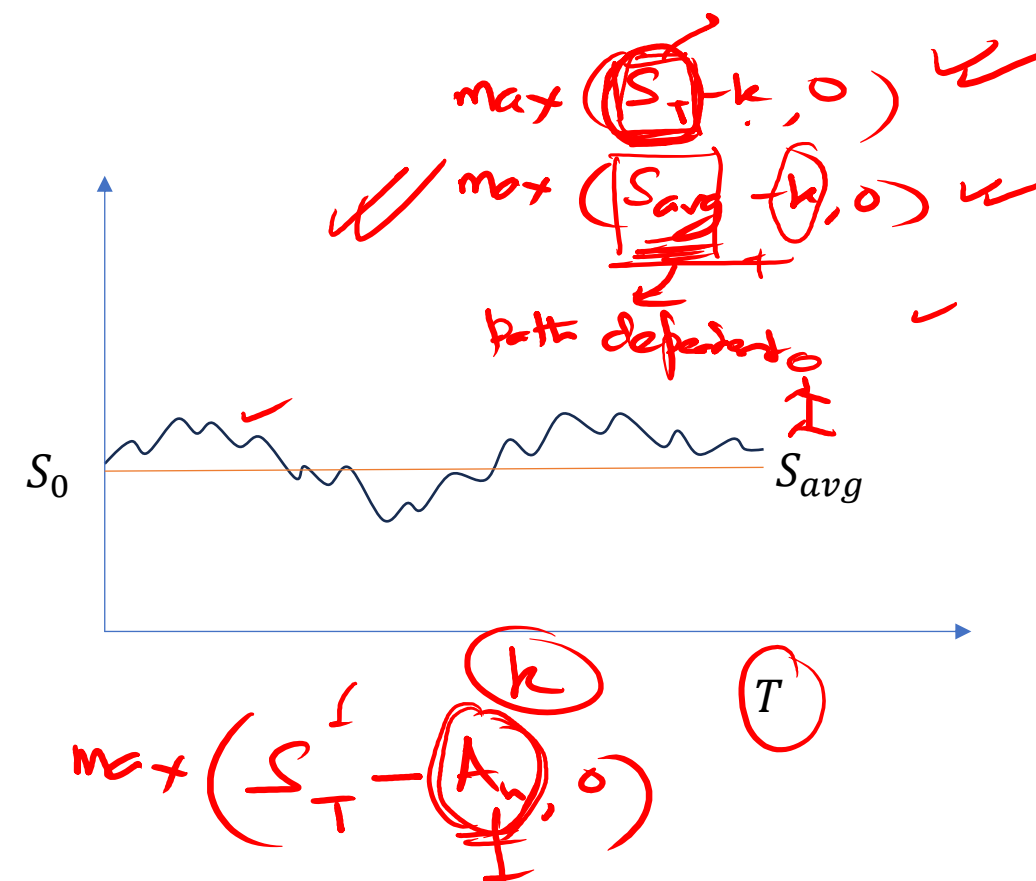
Asian Options

An **Asian option** is one whose payoff includes a time average of the underlying asset price. The Asian options can vary depending on the key dimensions below

- Sampling method (Continuous or Discrete)
- Averaging (Arithmetic or Geometric)
- Strike (Fixed or Floating)

$$(s_1, s_2, s_3, \dots, s_n)$$

$$A_n = \left(\frac{\sum s_i}{n} \right), \quad G_n = (s_1 s_2 \dots s_n)^{1/n}$$



Fixed Strike

Below are the specifications of different sampling and averaging schemes under Fixed Strike Call Option

Discrete Sampling

$$\rightarrow \max \left\{ \frac{1}{n} \sum_{i=1}^n S_{t_i} - K, 0 \right\} \quad (\text{Arithmetic})$$

$$\rightarrow \max \left\{ \prod_{i=1}^n (S_{t_i})^{\frac{1}{n}} - K, 0 \right\} \quad (\text{Geometric})$$

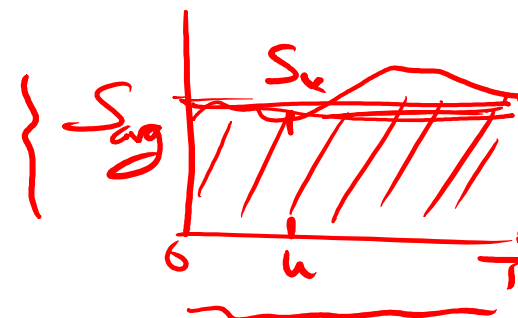
$$G = \prod_{i=1}^n (S_{t_i})^{\frac{1}{n}}$$

$$\rightarrow \log(G) = \frac{1}{n} [\log S_1 + \log S_2 + \dots + \log S_n]$$

Continuous Sampling

$$\max \left\{ \frac{1}{T} \int_0^T S_u du - K, 0 \right\} \quad (\text{Arithmetic})$$

$$\max \left\{ \exp \left(\frac{1}{T} \int_0^T \ln(S_u) du \right) - K, 0 \right\} \quad (\text{Geometric})$$



$$\text{Area} = \int_0^T S_u du$$

$$= T \times S_{\text{avg}}$$

$$\max \left[S_T - S_{\text{avg}}, 0 \right]$$

Floating Strike

For Floating Strike Call Option

Discrete Sampling

$$\max \left\{ \underline{S_T} - \frac{1}{n} \sum_{i=1}^n S_{t_i}, 0 \right\} \quad (\text{Arithmetic})$$

$$\max \left\{ \underline{S_T} - \prod_{i=1}^n (S_{t_i})^{\frac{1}{n}}, 0 \right\} \quad (\text{Geometric})$$

Continuous Sampling

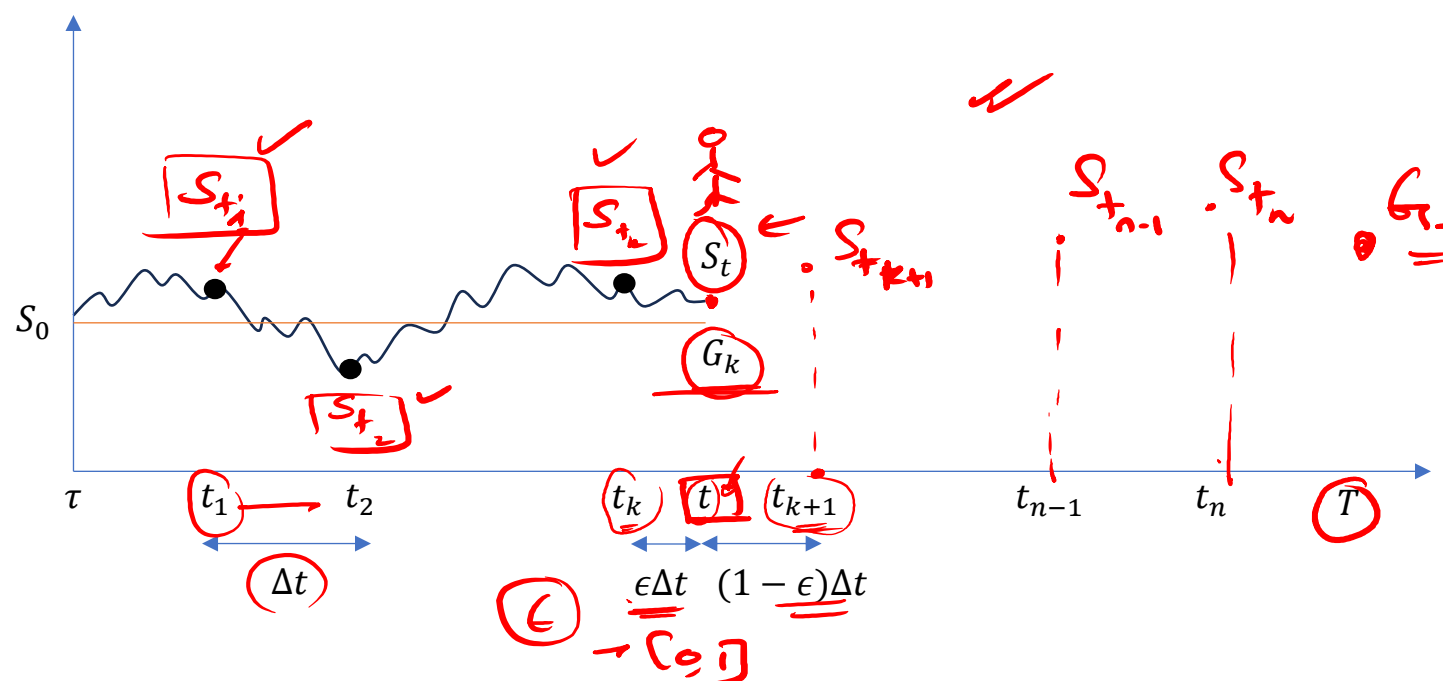
$$\max \left\{ \underline{S_T} - \frac{1}{T} \int_0^T S_u du, 0 \right\} \quad (\text{Arithmetic})$$

$$\max \left\{ \underline{S_T} - \exp \left(\frac{1}{T} \int_0^T \ln(S_u) du \right), 0 \right\} \quad (\text{Geometric})$$

Pricing the discrete geometric(I)

Asian options typically do not have closed form solutions except the one that uses Geometric average. Let's derive the analytical pricing formula for the same.

Consider the sampling window $[\tau, T]$ with sampling dates $(\tau) \leq \underline{t_1} < \underline{t_2} < \dots < \underline{t_n} \leq (T)$ and $S_{t_1}, S_{t_2}, \dots, S_{t_n}$ are the asset prices on the sampling dates. Suppose, we are currently at t , and the stock price is S_t . We have observed $S_{t_1}, S_{t_2}, \dots, S_{t_k}$.



Pricing the discrete geometric(II)

The final underlier is given as

$$\begin{aligned}
 \underline{G_n(\tau, T)} &= \left(\prod_{i=1}^n S_{t_i} \right)^{\frac{1}{n}} = \left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} \left(\prod_{i=k+1}^n S_{t_i} \right)^{\frac{1}{n}} = \left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} \underbrace{(S_{t_n} S_{t_{n-1}} S_{t_{n-2}} \dots S_{t_{k+2}} S_{t_{k+1}})^{\frac{1}{n}}} \\
 &\Rightarrow \left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} \underbrace{S_t^{\frac{n-k}{n}}}_{(1-\epsilon)\Delta t, t \rightarrow (k+1)} \left[\underbrace{\left(\frac{S_{t_n}}{S_{t_{n-1}}} \right)}_{(k+1)} \underbrace{\left(\frac{S_{t_{n-1}}}{S_{t_{n-2}}} \right)}_{(k+1)} \underbrace{\left(\frac{S_{t_{n-2}}}{S_{t_{n-3}}} \right)}_{(k+1)} \dots \underbrace{\left(\frac{S_{t_{k+1}}}{S_t} \right)}_{(k+1)} \right]^{\frac{1}{n}} \\
 &\quad \text{Handwritten notes: } (S_t^{n-k}), \eta - ? = (k+1), (?) = \eta - k - 1
 \end{aligned}$$

For $i = k+2, k+3, \dots, n-1, n$ under risk-neutral measure $\mathbb{Q}$,

$$\begin{aligned}
 \log\left(\frac{S_{t_i}}{S_{t_{i-1}}}\right) &= \left(r - \frac{1}{2}\sigma^2\right)\Delta t + \sigma \underbrace{(W_{t_i} - W_{t_{i-1}})}_{N(0, \Delta t)} \quad \text{and} \\
 \rightarrow \log\left(\frac{S_{t_{k+1}}}{S_t}\right) &= \left(r - \frac{1}{2}\sigma^2\right)\underbrace{(1-\epsilon)\Delta t}_{N(0, (1-\epsilon)\Delta t)} + \sigma(W_{t_{k+1}} - W_t)
 \end{aligned}$$

Pricing the discrete geometric(III)

Taking log

known

$$\log(G_n(\tau, T)) = \log \left[\left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} S_t^{\frac{n-k}{n}} \right] + \frac{1}{n} \left[\log \left(\frac{S_{t_n}}{S_{t_{n-1}}} \right) + 2 \log \left(\frac{S_{t_{n-1}}}{S_{t_{n-2}}} \right) + \dots + (n-k) \log \left(\frac{S_{t_{k+1}}}{S_t} \right) \right]$$

$$\underline{m} = \mathbb{E}^{\mathbb{Q}}[\log(G_n(\tau, T))] = \log \left[\left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} S_t^{\frac{n-k}{n}} \right] + \frac{1}{n} \left(r - \frac{1}{2} \sigma^2 \right) \Delta t (1 + 2 + 3 + \dots + (n-k-1) + (n-k)(1-\epsilon))$$

$$\underline{m} = \log \left[\left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} S_t^{\frac{n-k}{n}} \right] + \left(r - \frac{1}{2} \sigma^2 \right) \Delta t \left[\frac{(n-k-1)(n-k)}{2n} + \frac{(n-k)(1-\epsilon)}{n} \right]$$

$$\underline{s^2} = \mathbb{V}^{\mathbb{Q}}[\log(G_n(\tau, T))] = \frac{\sigma^2 \Delta t}{n^2} (1^2 + 2^2 + \dots + (n-k-1)^2 + (n-k)^2(1-\epsilon))$$

$$= \sigma^2 \Delta t \left(\frac{(n-k-1)(n-k)(2n-2k-1)}{6n^2} + \frac{(n-k)^2(1-\epsilon)}{n^2} \right)$$

Pricing the discrete geometric(IV)

Log of geometric average follows normal distribution,

$$\log(G_n(\tau, T)) \sim N(m, s^2)$$

So, the geometric average follows lognormal distribution just like the stock price itself. Because of this property, we can write down the call option price in BSM style

$$C_{\text{fixed strike}}^{(g)}(S_t, G_k, t; K, T) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[\max\{G_n(\tau, T) - K, 0\} | \mathcal{F}_t]$$

$$= e^{-r(T-t)} \left[e^{m + \frac{1}{2}s^2} \Phi(d_+) - K \Phi(d_-) \right]$$

$$\begin{matrix} \downarrow & \downarrow \\ m & s^2 \\ \uparrow & \uparrow \end{matrix}$$

$$d_+ = \frac{m - \ln K + s^2}{s}$$

$$d_- = d_+ - s \quad \sigma\sqrt{T}$$

$$d_+ = \frac{\log S_0 + (r - \frac{1}{2}\sigma^2)T - \log K + \sigma^2 T}{\sigma\sqrt{T}} = \frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$